

1865 jmbe Journal of Microbiology & Biology Education J Microbiol Biol Educ American Society for Microbiology (ASM) PMC11360556 11360556 11360556 38888314 10.1128/jmbe.00050-24 Educational activity of enzyme kinetics in an undergraduate biochemistry course: invertase enzyme as a model Al-Odat Ibrahim 1  Westernberg Dave J Editor 2 1 Department of Medical Laboratory Science, Faculty of Allied Medical Sciences, Jadara University, Irbid, Jordan 2 Missouri University of Science and Technology, Rolla, Missouri, USA  Address correspondence to Ibrahim Al-Odat, ibrahim.odat@jadara.edu.jo The author declares no conflict of interest. 18 6 2024 25 2 e00050-24 e00050-24 30 8 2024 Copyright © 2024 Al-Odat. This is an open-access article distributed under the terms of the Creative Commons

Attribution-NonCommercial-NoDerivatives 4.0 International license . ABSTRACT This article aims to simplify and facilitate the process of practical teaching of enzyme kinetics by utilizing minimal teaching laboratory requirements. Simultaneously, it ensures that students comprehend the enzyme kinetics experiment effectively. The focus is on teaching students how to estimate the maximum velocity (V_{max}) and Michaelis constant (K_m) of β -fructofuranosidase enzyme (also known as invertase) isolated from dry yeast. The invertase enzyme catalyzes the hydrolysis of sucrose substrate into glucose and fructose, employing the Michaelis-Menten approach of evaluating invertase enzyme kinetics as well as Lineweaver-Burk linear graphic approach of evaluating the Michaelis-Menten enzyme kinetics. The practical experiment seeks to reinforce the concepts of initial velocity dependence on substrate concentration. The data presented in the work were generated from a genuine practical biochemistry course enrolled by second-year undergraduate students in the Department of Pharmacy and the Department of Medical Laboratory Science. While there were minor variations in the invertase enzyme kinetic parameters among students, they successfully carried out the experiment. The students accurately estimated the V_{max} and K_m of the invertase enzyme in the sucrose hydrolysis chemical reaction. Moreover, they demonstrated an understanding of the meanings of the kinetic parameters (K_m and V_{max}) and the utility of the Lineweaver-Burk plot. KEYWORDS: education, practical biochemistry, invertase enzyme, enzyme kinetics, V_{max} and K_m status released display-pdf yes is-olb no is-manuscript no is-preprint no is-journal-matter no is-scanned no is-retracted no Received 2024 Mar 29; Accepted 2024 May 29; Collection date 2024 Aug.

INTRODUCTION Enzymology is a significant area of biochemistry (1). It encompasses the study of enzyme structure, function, regulation, and kinetics (2 , 3). Enzymes, specialized proteins serving as biological catalysts in living organisms (4), are a focal point in biochemistry education. Undergraduate practical biochemistry courses often introduce students to the basics of enzyme kinetics (5), emphasizing a hands-on approach for a better understanding of enzyme reactions and kinetics (4). However, incorporating enzyme kinetics experiments into undergraduate education poses challenges for many tertiary institutions, whether due to limited resource availability or inadequately equipped laboratories, particularly in newly developed settings. Nevertheless, the teaching methodology plays a crucial role in effectively conveying the practical objectives to undergraduate students (6). In such cases, refining the educational methodology of enzyme kinetic experiments becomes essential. This experimental tip addresses the challenge by providing affordable and readily available materials and instruments. Designed to be used with minimal hazards manageable through basic safety procedures, these resources facilitate the execution of the experiment in any teaching laboratory setting. Background This enzyme kinetics experiment utilized invertase enzyme as a model to determine the maximum velocity (V_{max}) and Michaelis constant (K_m) of the enzyme. Invertase enzyme, also known as β -fructofuranosidase, is a globular protein that catalyzes the hydrolysis of sucrose disaccharide (table sugar) into glucose and fructose (7). $Sucrose + H_2O \xrightarrow{invertase} glucose + fructose$ Invertase is naturally produced in various plants and microorganisms, including *Saccharomyces cerevisiae* (8). In this practical procedure, students track the sucrose hydrolysis reaction facilitated by invertase enzyme by measuring the concentration of glucose produced in the reaction tubes. The reaction conditions of invertase enzyme hydrolysis of sucrose in this procedure were set at a temperature of 30°C and a neutral pH. The reaction is allowed to proceed for a time interval of 20 min. Students measured the glucose concentration at different time points using a glucometer. Intended audience The current tip seeks to streamline the instruction of enzyme kinetics in a practical experiment to first- or second-year undergraduate students of allied health majors as well as non-major biochemistry undergraduate students from diverse backgrounds. Students in this level normally have no previous experience and require laboratory skills. The objective of this experiment is to facilitate a straightforward and comprehensive understanding of the enzyme kinetics topic. However, by the end of this activity, students should be able to define the different terms of enzyme kinetics. In addition, students should be able to estimate the kinetic parameters V_{max} and K_m from enzyme saturation data using direct fitting and Lineweaver-Burk methods. The approach involves employing invertase enzyme as a model in the experiment to estimate both V_{max} and K_m , using the Lineweaver-Burk method to evaluate Michaelis-Menten kinetics. PROCEDURE Safety issues These lab procedures adhere to the recommendations set by the ASM Guidelines for Biosafety in Teaching Laboratories. No hazardous materials were involved in this experiment; however, students were advised to wear personal protective equipment and adhere to safety measures throughout the procedure, as outlined in the orientation session conducted at the onset of their undergraduate studies. Materials and chemicals The following materials and chemicals are required in order to perform the experiment. Six test tubes and a rack Graduated cylinders and beakers Spatula 0.4 M sucrose stock solution Dry yeast Distilled water Different ranges of automatic micropipettes and tips Microscopic slides and disposable droppers Glucometer and strips 30°C Water bath Timer Experimental methodology Step-by-step students' instructions are provided in the worksheet supplementary file number (1) (Data not shown). Preparing of invertase enzyme solution (Appendix: Part A) The invertase enzyme solution is prepared in advance by the technical staff. This was done by weighing 0.25 g of dry yeast into a 500-mL glass beaker. The dry yeast was suspended in 250 mL warm distilled water (30°C). The suspension was left in the water bath of 30°C for 20 min with periodic stirring. The suspension was stored at 30°C in the water bath to serve as an invertase enzyme solution. Substrate solutions (Appendix: Part B) Students use automatic micropipettes to prepare various concentrations of sucrose substrate solutions by diluting a 0.4 M sucrose stock solution, as detailed in Table 1 . TABLE 1 Sucrose substrate solutions with various concentrations Tube # d. H_2O (mL) Sucrose solution volume (mL) Final concentration (M) 1 1 1 mL from 0.4 M sucrose stock solution 0.2

2 1 1 mL from tube 1 0.1 3 1 1 mL from tube 2 0.05 4 1 1 mL from tube 3 0.025 5 1 1 mL from tube 4 0.0125 6 1 1 mL from tube 5 0.00625

Preparing the reactions (Appendix: Part C) Substrate solutions are pre-incubated in a warm water bath at 30°C for 10 min to bring the solutions to the reaction temperature. To initiate the reactions, while the substrate solutions are in the water bath, students use an automated micropipette to add 1 mL of the invertase enzyme solution to each substrate reaction tube at a specific time, as outlined in Table 2. This time point marks the start of the chemical reaction of sucrose hydrolysis by invertase enzyme. TABLE 2 Time point of starting the invertase chemical reaction Elapsed time (min) Tube # Enzyme volume 10 1 1 mL 11 2 1 mL 12 3 1 mL 13 4 1 mL 14 5 1 mL 15 6 1 mL Reading glucose concentration (Appendix: Part D) After 20 min from the initiation of the chemical reactions, the concentration of glucose produced in the reaction tubes due to sucrose hydrolysis by invertase enzyme is measured. Using a glucometer and glucometer strips, one drop from each reaction tube is placed on a microscopic slide. A glucometer strip was inserted into the glucometer and the strip was touched to the reaction solution on the microscopic slide in order to read the glucose concentration. The glucometer readings were recorded as per the time points outlined in Table 3. TABLE 3 Time point for reading glucose concentration in the reaction tubes Elapsed time (min) Tube # Enzyme volume Glucometer reading (mg/dL) 30 1 1 mL 674 31 2 1 mL 537 32 3 1 mL 425 33 4 1 mL 288 34 5 1 mL 198 35 6 1 mL 162 CONCLUSION

The V_o of the invertase enzyme in each reaction tube, representing a specific sucrose concentration, was calculated based on the glucose concentration readings provided in Table 4. TABLE 4 Average V_o calculation in the reaction tubes Tube # [Sucrose] ($\mu\text{mol/mL}$) Average [glucose] (mg/dL) [Glucose] (mmol/L) = mg/dL $\times 0.0555$ [Glucose] ($\mu\text{mol/mL}$) = $\mu\text{mol/mL}$ Average V_o ($\mu\text{mol/min/mL}$) = $\mu\text{mol} / \text{min} / \text{mL}$ 20 1 2×10^{-4} 674 1.87×10^{-4} 537 1.49×10^{-4} 425 1.18×10^{-4} 288 0.80×10^{-4} 198 0.55×10^{-4} 162 0.45×10^{-4}

To estimate the V_{max} and K_m of the invertase enzyme from the Michaelis-Menten curve (Fig. 1) using the Excel application, students Plot the sucrose concentrations on the x-axis and the reaction velocity on the y-axis. They estimated the V_{max} from the y-axis value corresponding to the flat part of the curve. The K_m is estimated from the sucrose concentration on the x-axis corresponding to half of the V_{max} . Fig 1 Michaelis-Menten curve of invertase enzyme. The curve exhibits the expected hyperbolic shape consistent with the Michaelis-Menten kinetic mechanism, reflecting the influence of substrate concentration on enzyme kinetics. However, accuracy in estimating enzyme kinetics may be compromised due to the continuous increase in reaction velocity in the flat part of the curve. To estimate the V_{max} and K_m of the invertase enzyme from the Lineweaver-Burk curve (Fig. 2) using Microsoft Excel, students plot the inverse of sucrose concentrations on the x-axis and the inverse of reaction velocity on the y-axis. Then they estimated the V_{max} from the y-intercept (equal to $1/V_{\text{max}}$), and K_m is estimated from the x-intercept (equal to $-1/K_m$) of the generated straight line. Fig 2 Lineweaver-Burk plot of invertase enzyme. $y = 0.0003x + 0.0015$. $R^2 = 0.9893$. This enzyme kinetic practical experiment introduced the students to the basics of enzyme kinetics in an easy manner. Students were able to comprehend the different terms of enzyme kinetics through hands-on experiments and were able to estimate the V_{max} and K_m of invertase enzyme from the Michaelis-Menten curve and Lineweaver-Burk plot. These achievements were evident in the laboratory report the students submitted to the instructor in the following week. Individual reports were required by every student which include the V_{max} and K_m values estimated from the Michaelis-Menten curve and Lineweaver-Burk plot constructed manually on a graph paper or by spreadsheet. All reports were marked and corrected to maximize students' learning outcomes. Upon reflecting from students, they seem to have gained the understanding of the basic enzyme kinetics and were able to use spreadsheet to estimate the V_{max} and K_m of invertase enzyme employed in the experiment. Many students scored high marks in the lab report assessment. ACKNOWLEDGMENTS I thank all the technical staff in the Department of Pharmacy and the Department of Medical Laboratory Science at Jadara University for their assistance in preparing for these practical sessions over three academic semesters. I also extend my thanks to the students from both departments enrolled in Pharmacy and Medical Laboratory Science programs at Jadara University. Their contributions were invaluable to the generation of the data used in the analysis presented in the paper, during the fall and spring semesters of the academic year 2021/2022 and the fall semester of the academic year 2022–2023. I declare that no funds, grants, or other support was received during the preparation of the manuscript. Contributor Information Ibrahim Al-Odat, Email: ibrahim.odat@jadara.edu.jo. Dave J. Westenberg, Missouri University of Science and Technology, Rolla, Missouri, USA SUPPLEMENTAL MATERIAL The following material is available online at <https://doi.org/10.1128/jmbe.00050-24>. Supplemental Material [jmbe.00050-24-s0001.docx](#) Supplemental appendix (laboratory worksheet: enzyme kinetics). 10.1128/jmbe.00050-24.SuF1 ASM does not own the copyrights to Supplemental Material that may be linked to, or accessed through, an article. The authors have granted ASM a non-exclusive, world-wide license to publish the Supplemental Material files. Please contact the corresponding author directly for reuse. REFERENCES 1. El-Khateeb A. 2020. Practical biochemistry principles and techniques approach. *Prog Chem Biochem Res* 3:180–193. doi: 10.33945/SAMI/PCBR.2020.2.9 2. González-Cruz J, Rodríguez-Sotres R, Rodríguez-Penagos M. 2003. On the convenience of using a computer simulation to teach enzyme kinetics to undergraduate students with biological chemistry-related curricula. *Biochem Molecular Bio Educ* 31:93–101. doi: 10.1002/bmb.2003.494031020193 3. Srinivasan B. 2021. Words of advice: teaching enzyme kinetics. *FEBS J* 288:2068–2083. doi: 10.1111/febs.15537 4. Leksmono CS, Manzoni C, Tomkins JE, Lucchesi W, Cottrell G, Lewis PA. 2018. Measuring lactase enzymatic activity in the teaching lab. *J Vis Exp* 138:e54377. doi: 10.3791/54377 5. Miquet JG, González L, Sotelo AI, González Lebrero RM. 2019. A laboratory work to introduce biochemistry undergraduate students to basic enzyme kinetics: alkaline phosphatase as a model. *Biochem Mol Biol Educ* 47:93–99. doi: 10.1002/bmb.21195 6. I. Elzagheid M. 2018. Laboratory activities to introduce carbohydrates qualitative analysis to college students. *World J Chem Educ* 6:82–86. doi: 10.12691/wjce-6-2-1 7. Manoochehri H, Hosseini NF, Saidijam M, Taheri M, Rezaee H, Nouri F. 2020. A review on invertase: its potentials and applications. *Biocatal Agric Biotechnol* 25:101599. doi: 10.1016/j.bcab.2020.101599 8. Sainz-Polo MA, Ramírez-Escudero M, Lafraya A, González B, Marín-Navarro J, Polaina J, Sanz-Aparicio J. 2013. Three-dimensional structure of *Saccharomyces* invertase: role of a non-catalytic domain in oligomerization and substrate specificity. *J Biol Chem* 288:9755–9766. doi: 10.1074/jbc.M112.446435 Associated Data Supplementary Materials Supplemental Material [jmbe.00050-24-s0001.docx](#) Supplemental appendix (laboratory worksheet: enzyme kinetics).

